

Types of Operators

Single-Valued

$$T: X \rightarrow Y$$

Each x maps to exactly one $tx \in Y$

Set-Valued

$$A: X \rightarrow 2^Y$$

Each x maps to a set $Ax \subseteq Y$

Graph and Inverse

$$\text{Graph: } \text{gra } A = \{(x, y) \in X \times Y : y \in Ax\}$$

$$\text{Inverse: } A^{-1} = \{(y, x) \in Y \times X : (x, y) \in \text{gra } A\}$$

$$\text{Domain: } \text{dom } A = \{x \in X : Ax \neq \emptyset\}$$